

Experiment No. 5

Aim

Socket programming for transport layer packet (UDP client)

Theory

The **UDP Client** operates as an active sender in a connectionless architecture. Because UDP does not maintain a persistent connection, the client can immediately transmit datagrams to a target server once the destination IP address and port number are known.

UDP Client Execution Steps:

1. **Socket Creation (socket()):** The client creates a datagram socket using AF_INET (IPv4) and SOCK_DGRAM (UDP).
2. **Addressing Setup:** The client configures a target address structure (sockaddr_in) containing the remote server's IP address and port number.
3. **Data Transmission (sendto()):** The client sends data directly using sendto(). Since UDP is connectionless, no connect() call is required.
4. **Data Reception (recvfrom()):** The client optionally waits for a reply from the server using recvfrom().
5. **Closing Socket (close()):** Closes the client socket descriptor upon completion.

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```
lab-03-25@lab-03-25-OptiPlex-3280-AIO:~$ vim client_udp284.c
lab-03-25@lab-03-25-OptiPlex-3280-AIO:~$ gcc client_udp284.c -o client
lab-03-25@lab-03-25-OptiPlex-3280-AIO:~$ ./client
Enter message: hello
Server Response: Hello from UDP Server
lab-03-25@lab-03-25-OptiPlex-3280-AIO:~$ ./client
Enter message: tanishq 284
Server Response: Hello from UDP Server
lab-03-25@lab-03-25-OptiPlex-3280-AIO:~$
```

```

#include <stdio.h>
#include <string.h>
#include <unistd.h>
#include <arpa/inet.h>
int main()
{
    int sockfd;
    char message[100], buffer[100];
    struct sockaddr_in serverAddr;
    socklen_t addr_size;
    // Create UDP socket
    sockfd = socket(AF_INET, SOCK_DGRAM, 0);
    if (sockfd < 0)
    {
        printf("Socket creation failed\n");
        return 1;
    }
    // Configure server address
    serverAddr.sin_family = AF_INET;
    serverAddr.sin_port = htons(8080);
    serverAddr.sin_addr.s_addr = inet_addr("127.0.0.1");
    printf("Enter message: ");
    fgets(message, sizeof(message), stdin);
    // Send message to server
    sendto(sockfd, message, strlen(message), 0,
    (struct sockaddr *)&serverAddr, sizeof(serverAddr));
    // Receive response from server
    addr_size = sizeof(serverAddr);
    int n = recvfrom(sockfd, buffer, sizeof(buffer) - 1, 0,
    (struct sockaddr *)&serverAddr, &addr_size);
    buffer[n] = '\0';
    printf("Server Response: %s\n", buffer);

    return 0;
}

```

Conclusion:

The UDP Client socket was successfully constructed at the Transport Layer. Using SOCK_DGRAM and the sendto() system call, the client demonstrated connectionless communication by sending datagram packets directly to the server without needing to perform a TCP-style connection handshake.